Assignment 1:

Write a C program that includes a user-defined function named isPrime with the signature int isPrime(int num); The function should take an integer as a parameter and return 1 if the number is prime and 0 otherwise.

Source Code:

```
#include<stdio.h>
#include<math.h>

int isPrime(int num){
   if (num<=1)
      return 0;
   for (int i=2;i<=sqrt(num);i++) {
      if (num%i==0)
         return 0;
   }
   return 1;
}

int main() {
   int n;
   printf("Enter the number : ");
   scanf("%d", &n);
   if (isPrime(n))
      printf("This is a prime number.");
   else
      printf("This is not a prime number.");
   return 0;
}
```

```
Enter the number :23

This is a prime number.
Process finished with exit code 0
```

```
Enter the number :32

This is not a prime number.
Process finished with exit code 0
```